Dystric Nitisols:

Dystric Nitisols are deep, well-drained tropical soils characterized by a reddish color due to iron oxide accumulation and a clayey subsurface (nitic horizon) with strong, polyhedric blocky structure. They are named "Dystric" because of their low base saturation (<35%), making them acidic and less fertile compared to other Nitisols. Despite this, they are physically stable, with a porous, deep solum that supports deep rooting.

Drainage:

- **Wet season:** Well-drained; water infiltrates easily due to stable structure.
- **Dry season:** Soil remains friable and workable; moderate moisture retention supports crops.
- **Landscape:** Common on undulating to hilly tropical terrain; rarely found in depressions.

Depth:

- Very deep (>150 cm).
- Rooting depth is generally unrestricted due to stable structure, although low nutrient availability may limit growth.

Workability:

- **When wet:** Plastic and sticky but manageable; does not smear like Vertisols.
- **When dry:** Firm and friable; easy to till and prepare seedbeds

Nutrient Richness & Cation Exchange Capacity (CEC):

- CEC: Moderate to high (15–35 cmol(+)/kg).
- Low base saturation (<50%), making them acidic.
- Low in available phosphorus and calcium; nitrogen and organic matter are low.
- Iron oxides dominate; some micronutrients (zinc, manganese) may also be low.

pH:

- Typically acidic (pH 5.0–6.5).

Management:

- Highly productive soils if well managed; their deep, porous solum and stable structure permit deep rooting and resistance to erosion.
- Good workability, internal drainage, and fair water-holding capacity support both plantation and food crops.
- Commonly planted with cocoa, coffee, rubber, pineapple, and food crops on smallholdings.
- High phosphorus sorption requires P-fertilizer application: slow-release, low-grade rock phosphate (several tons/ha every few years) combined with small doses of soluble superphosphate for short-term crop response.
- Maintain vegetation cover or cover crops to reduce erosion on slopes.
- Incorporate lime to correct acidity and improve base saturation.